EduPulse AWS Cost Analysis & Reduction Plan (Revised)

> [!IMPORTANT] Target: **$60–70/month**. All estimates use **us-east-1** on-demand pricing (Nov 2026). Actual costs depend on your exact region, task sizes, and traffic. ✅ Corrected: no NAT Gateway — both services use `AssignPublicIp: ENABLED`.

✅ Corrected Architecture Facts

Item	Reality
App Service (`AppService`)	`AssignPublicIp: DISABLED` → runs in **private subnets**
Scheduler Service (`SchedulerService`)	`AssignPublicIp: ENABLED` → runs in **public subnets**
NAT Gateway	**You confirmed: none ✅**
App task default CPU/mem	Workflow fallback: `512` CPU / `1024` MB
Scheduler task default CPU/mem	Workflow fallback: `256` CPU / `512` MB
Secrets	11 secrets in Secrets Manager
Log groups	4 groups, 7-day default retention
ECR	2 repos, 5-image retention
Scheduler desired count	Defaults to `0` ✅

> [!WARNING] **Important:** `AppService` has `AssignPublicIp: DISABLED` and uses `PrivateSubnetIds`. Without a NAT Gateway, this task **cannot reach the internet** (ECR image pulls, Secrets Manager, SES, Paystack). It must be relying on **VPC Interface Endpoints** or be misconfigured. Check this — if it's actually working, you have VPC endpoints (which cost ~$7–10/mo each). If it's broken, switching to public subnets saves both the endpoint cost AND the fix.

🔍 Cost Breakdown — Current vs. Optimized

Service	What you have	Est. $/mo now	Optimized config	Est. $/mo after
ALB	1 internet-facing ALB	~$18–22	Same (unavoidable)	~$18
ECS Fargate – App	1 task, 512 CPU / 1024 MB (default)	~$18	256 CPU / 512 MB	~$9
ECS Fargate – Scheduler	0 tasks ✅	$0	Keep at 0	$0

Service	What you have	Est. $/mo now	Optimized config	Est. $/mo after
VPC Endpoints (if any)	Unknown — needed if app is in private subnet w/o NAT	~$7–28	Eliminate by moving app to public subnet	$0
Database (RDS)	External — unknown instance size	$15–50	t3.micro, single-AZ	~$15
Secrets Manager	11 secrets × $0.40	~$4.40	Move to SSM Parameter Store	~$0
CloudWatch Logs	4 groups, 7-day retention	~$2–4	3-day retention	~$1
ECR	2 repos, 5-image limit	~$0.50	3-image limit	~$0.20
Route53	1 hosted zone + 2 records	~$0.50	Same	~$0.50
S3	Versioned document bucket	~$0.10	Same	~$0.10
ACM	1 certificate	Free	Free	$0
SNS + Alarms	2 alarms, 1 topic	~$0.10	Same	~$0.10
SES	Transactional email	~$0.10/1k	Same	~$0.10

🚨 The Real Cost Drivers (Corrected)

#1 — 🔴 App Task in Private Subnet with No NAT (Critical)

Your `AppService` is in `PrivateSubnetIds` with `AssignPublicIp: DISABLED` but you say there's no NAT Gateway. For this to work you must have **VPC Interface Endpoints** for:

- ECR (image pulls): `ecr.api` + `ecr.dkr` + `s3` gateway = ~$14–21/mo
- Secrets Manager: ~$7/mo
- CloudWatch Logs: ~$7/mo

That's up to $35/mo in hidden endpoint costs.

The fix is simple: Move the app task to **public subnets** (`AssignPublicIp: ENABLED`), just like the scheduler already does. This:

- Eliminates VPC endpoint costs entirely ($0 vs. ~$28–35/mo)
- Uses the free Internet Gateway for outbound traffic
- Is exactly what your Scheduler already does successfully

#2 — 🔴 ALB — $18–22/mo (Fixed, Unavoidable)

The ALB is `\$0.008/hr` base + LCU charges. It's unavoidable while you need HTTPS. Accept it as a fixed baseline.

#3 — 🟡 App Task Over-sized — Saves ~$9/mo

Your workflow defaults to `ECS_APP_CPU=512` / `ECS_APP_MEMORY=1024` (0.5 vCPU / 1 GB). Dropping to `256`/`512` halves the Fargate compute cost.

Config	$/mo (1 task, 24/7)
512 CPU / 1024 MB (current default)	~$18
256 CPU / 512 MB ✅	**~$9**

#4 — 🟡 Secrets Manager — $4.40/mo

11 secrets × $0.40/mo = $4.40, plus API call charges on every container start. **SSM Parameter Store SecureString is free** for standard parameters.

#5 — 🟢 Database (Unknown)

Your `DATABASE_URL` is external. If it's RDS:

- **t3.micro, single-AZ** = ~$15/mo ✅ fine
- **t3.small or higher** = $30+/mo → downsize

✅ Action Plan to Hit $60–70/mo

Step 1 — Fix the App Service Network Config (Biggest Win)

Move `AppService` from private to public subnets in the CloudFormation template:

```
# In AppService NetworkConfiguration — change to:
AwsvpcConfiguration:
  AssignPublicIp: ENABLED          # was DISABLED
  SecurityGroups:
    - !Ref AppServiceSecurityGroup
  Subnets: !Ref PublicSubnetIds    # was PrivateSubnetIds
```

Then delete any VPC Interface Endpoints you have for ECR, Secrets Manager, and CloudWatch.

Step 2 — Right-size the App Task

Set in GitHub Actions environment variables:

```
ECS_APP_CPU=256
ECS_APP_MEMORY=512
GUNICORN_WORKERS=2
```

Step 3 — Migrate Secrets Manager → SSM Parameter Store (saves $4.40/mo)

Create SSM SecureString parameters to replace all 11 Secrets Manager secrets:

```
# Example — repeat for each secret
aws ssm put-parameter \
  --name /edupulse/production/database-url \
  --type SecureString \
  --value "your-db-url"
```

Then update the task definition `secrets` block from `secretsmanager` ARNs to `ssm` ARNs:

```
// Before (Secrets Manager):
{"name": "DATABASE_URL", "valueFrom": "arn:aws:secretsmanager:..."}

// After (SSM — free):
{"name": "DATABASE_URL", "valueFrom":
"arn:aws:ssm:REGION:ACCOUNT:parameter/edupulse/production/database-url"}
```

The ECS task execution role needs `ssm:GetParameters` and `kms:Decrypt` (for the default SSM key). Update the IAM policy accordingly.

Step 4 — Lower Log Retention and ECR Retention

Update CloudFormation parameters:

```
LogRetentionInDays=3
EcrImageRetentionCount=3
```

Step 5 — Confirm Scheduler is at 0

```
ECS_SCHEDULER_DESIRED_COUNT=0   # (already the default, just confirm)
```

📊 Projected Monthly Cost After All Steps

Service	After $/mo
ALB	$18
ECS Fargate App (256 / 512 MB)	$9
ECS Fargate Scheduler	$0
VPC Endpoints (eliminated)	$0
RDS t3.micro single-AZ	$15
SSM Parameter Store (11 params)	$0

Service	After $/mo
CloudWatch Logs (3-day)	$1
ECR (2 repos, 3 images)	$0.20
Route53	$0.50
S3 (documents)	$0.10
SES	~$0.10
SNS + Alarms	$0.10
Total	**~$44–45**

> [!NOTE] Even adding buffer for data transfer, Fargate surge, and SES volume, the final bill should land solidly in the **$55–65/month** range — right in your target.

🗺️ Check These in Your AWS Console Now

1. **VPC Console** → **Endpoints** — count how many interface endpoints you have; each is ~$7/mo
2. **ECS Console** → **App task** — confirm actual CPU/Memory of the running task definition
3. **Cost Explorer** → **Group by Service** — check for any surprise services (ElastiCache, EC2, etc.)
4. **RDS Console** → confirm instance type and Multi-AZ is OFF
5. **Secrets Manager** → count your secrets (should be 11 — confirm none are free-tier grandfathered)